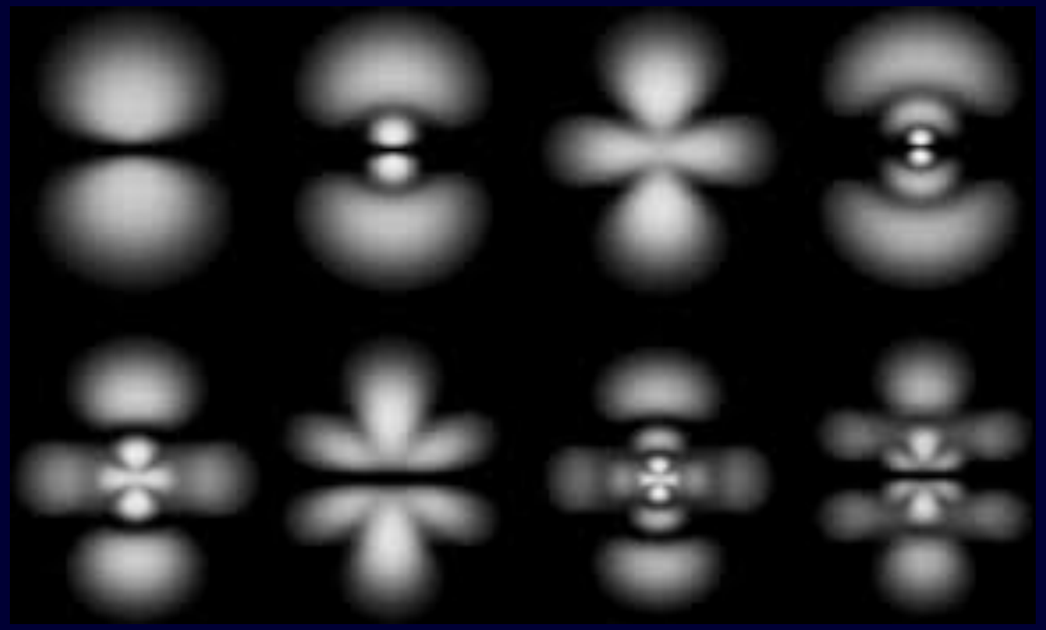
Chapters: §38:6; 39:1



Modern Physics: Part 7

SUZIE SHEEHY

SUMMER SEMESTER

Recap from last lecture...

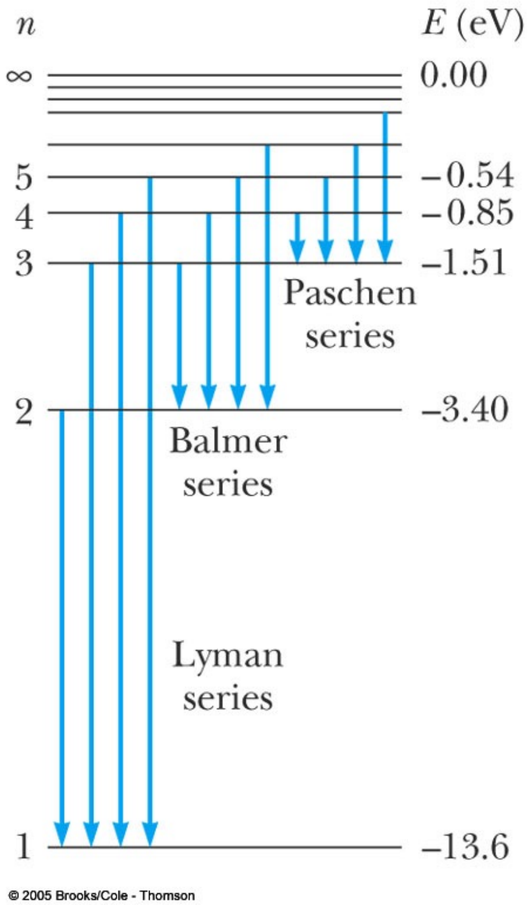
- De Broglie hypothesis: all matter has a frequency and wavelength

$$\lambda = \frac{h}{p}$$ is the de Broglie wavelength

- Bohr model of the atom: can calculate energy levels (for simple atoms)
- Emission and absorption spectra
- General formula for light emission by atoms

$$f = cR_H\left(\frac{1}{n_2^{\,2}} - \frac{1}{n_1^{\,2}}\right),$$

n_1, n_2 integers with $n_1 > n_2$



Objectives:

- Explain the relation between the de Broglie waves and energy quantization and and Bohr's quantization condition
- Identify that matter waves are described by the Schrödinger equation
- Understand the relation between de Broglie waves and matter wave packet
- Calculate the size of the Hydrogen atom

The Correspondence Principle §39.5

- Bohr put forward the idea that the *average* behavior of a quantum system looks like a classical one in the limit of the principal quantum number, $n \rightarrow \infty$.

- The radius of the Bohr hydrogen atom is $r = n^2 a_B$, the atom becomes a macroscopic object as n becomes very large.

- Bohr's idea, that the quantum world blends smoothly into the classical world for $n \rightarrow \infty$, is known as the **correspondence principle.**

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Correspondence Principle

"In the limiting case of $n \longrightarrow \infty$, the results obtained form quantum theory must agree with those obtained form classical theory".

e.g. Freq of EM radiation emitted form a highly excited H atom:
When the electron is far from the proton--say, a meter--it should act like a classical charge, emitting light with the same frequency that it orbits the proton.

Quantum theory:

Recall: E levels are close together and high n

$$f_{QT} = \frac{me^4}{4\pi(4\pi\varepsilon_0)^2\hbar^3}\left(\frac{1}{(n-1)^2} - \frac{1}{n^2}\right)$$

N → n-1 in small steps.
≈ cont. process.

$$= \frac{me^4}{4\pi(4\pi\varepsilon_0)^2\hbar^3}\frac{2n-1}{n^2(n-1)^2}$$

$$\approx \frac{me^4}{4\pi(4\pi\varepsilon_0)^2\hbar^3}\frac{2}{n^3}$$

for large n.
-- according to Quantum theory

Semi-Classical Theory:

$$f_{SCT} = \text{freq. of electron around nucleus}$$

$$= \frac{v}{2\pi\, r}$$

Using: $v_n = c\left(\frac{e^2}{4\pi\varepsilon_0 \hbar c}\right)\frac{1}{n}$ and $r_n = \frac{4\pi\varepsilon_0 n^2 \hbar^2}{me^2}$

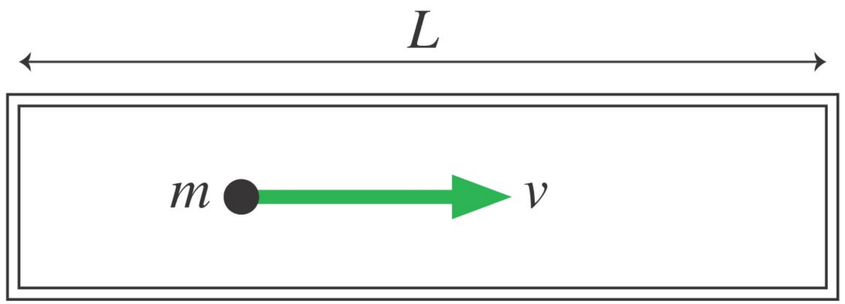
we obtain the semi-classical result:

$$f_{SCT} = \frac{me^4}{4\pi\left(4\pi\varepsilon_0\right)^2 \hbar^3}\frac{2}{n^3}$$

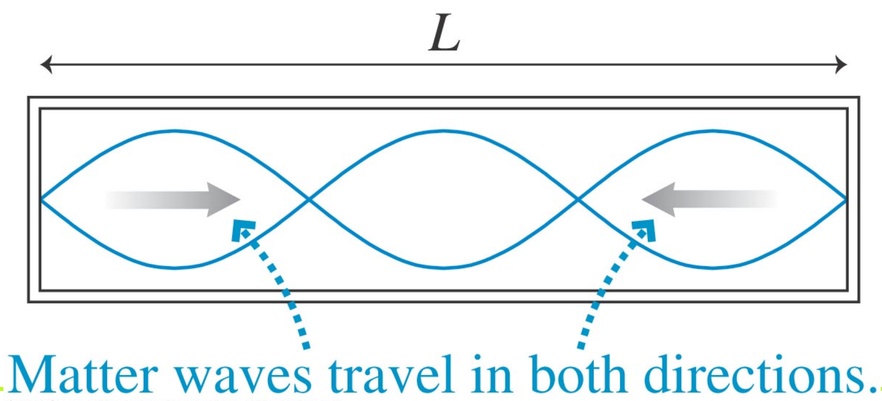
which is the same as the quantum-theory result in the large n limit.

De Broglie waves and energy quantization §39.1

Consider a matter wave confined to a small region of space, such as a particle (e.g. an electron) confined to a box.



Particle behavior:
- particle bounces and and forth:



Wave behavior:
- standing de Broglie wave in the box

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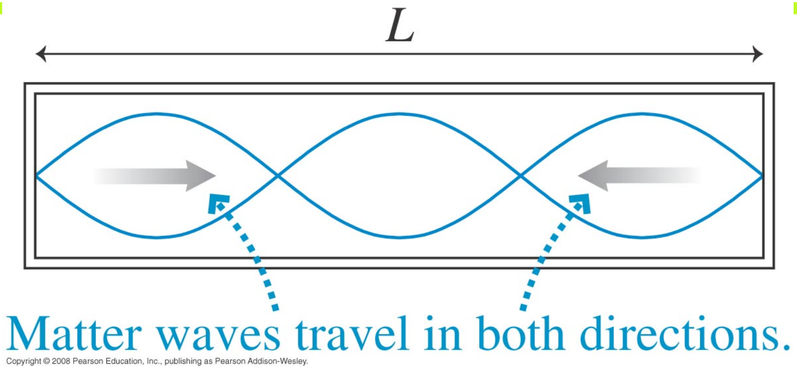

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A standing wave of length *L must* have a wavelength given by

$$\lambda = \frac{2L}{n} \qquad n = 1,2,3,\ldots$$

We must also satisfy the de Broglie wavelength equation

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

Equating the two expression for the wavelength, we find

$$v_n = n\left(\frac{h}{2mL}\right) \qquad n = 1,2,3$$

Only certain speeds (and thus certain kinetic energies) are allowed!

The particle's kinetic energy, is

$$E_n = \frac{1}{2} m v_n^2 = n^2 \frac{h^2}{8mL^2} \qquad n = 1,2,3$$

De Broglie's hypothesis about the wave-like properties of matter leads to the remarkable conclusion that **the energy of a confined particle is quantized.**

The allowed values of the energy are called energy levels.

The integer, *n*, that characterizes the energy levels is called the **principal quantum number.**

Calculate lowest energy level

- An electron in a 1D infinitely deep potential energy well of L=100pm
  - (a) what is the smallest amount of energy the electron can have?

$$\frac{h^2}{8mL^2} = \frac{(6.63 \times 10^{-34}\ J\,.\,s)^2}{(8)(9.11 \times 10^{-31}\ kg)(100 \times 10^{-12}\ m)^2}$$

- Calculate collection of constants $\frac{h^2}{8mL^2} = 6.031 \times 10^{-18}\ J$

$$\frac{h^2}{8mL^2}n^2 = (6.031 \times 10^{-18}\ J)(1^2)$$

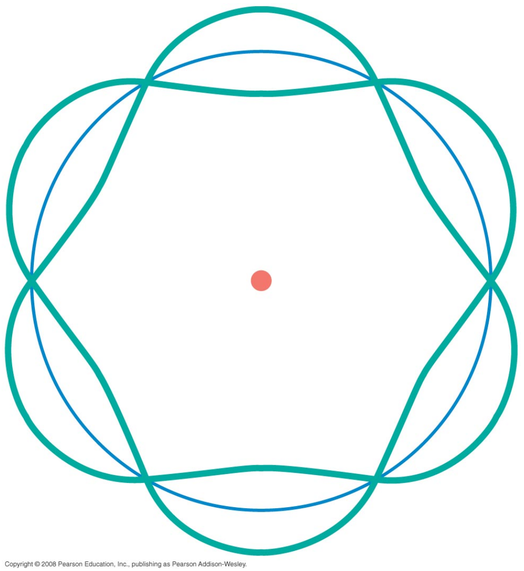
- Is the lowest energy level E_1, which is 37.3 eV

De Broglie waves and Bohr's quantization condition

Consider an electron wave going in a circle around a nucleus.

For the wave to be single-valued, we need:

Standing wave: Circumference → integer number of wavelengths



$$2\pi\, r = n\,\lambda = n\frac{h}{p}$$

$$\Rightarrow rp = n\frac{h}{2\pi} = n\,\hbar$$

But $rp = \left|\vec{L}\right|$ angular momentum

$$\Rightarrow \quad L = n\,\hbar$$

Which is Bohr's quantization condition!

de Broglie waves provide a physical picture for why the Bohr model works.

Electrons in an atom have no trajectory

For a well-defined trajectory we require:

Uncertainty in momentum in any direction << magnitude of momentum

Now $|p| = mv$ with $v = \frac{\hbar}{mr}$ (using $L = n\hbar = mvr$)

$\Rightarrow |p| = \frac{\hbar}{r}$ and thus $\Rightarrow \Delta p << \frac{\hbar}{r}$ (i)

For a well-defined trajectory we also require:

Uncertainty in position in any direction << radius of orbit

i.e. $\Delta y << r$ (ii)

Multiplying (i) and (ii) ➔ $\Delta y \Delta p_y << \hbar$

Which is in contradiction with the uncertainty principle.

➔No trajectory!

Estimate the Size of Hydrogen

Why can't an electron be at r =0? because of the uncertainty relation.

$$E = \frac{p^2}{2m} - \frac{Ze^2}{4\pi\varepsilon_0 r}$$

Kinetic Potential

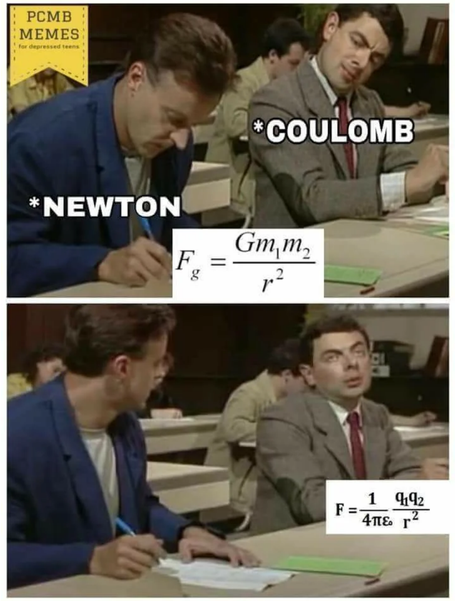
$$\Delta r \Delta p \sim \hbar$$

$$r \rightarrow \Delta r \qquad p \rightarrow \Delta p \sim \frac{\hbar}{\Delta r}$$

Let's find delta r: the distance of the electron from zero, assuming the minimum energy state.

Assume small deviations from zero.

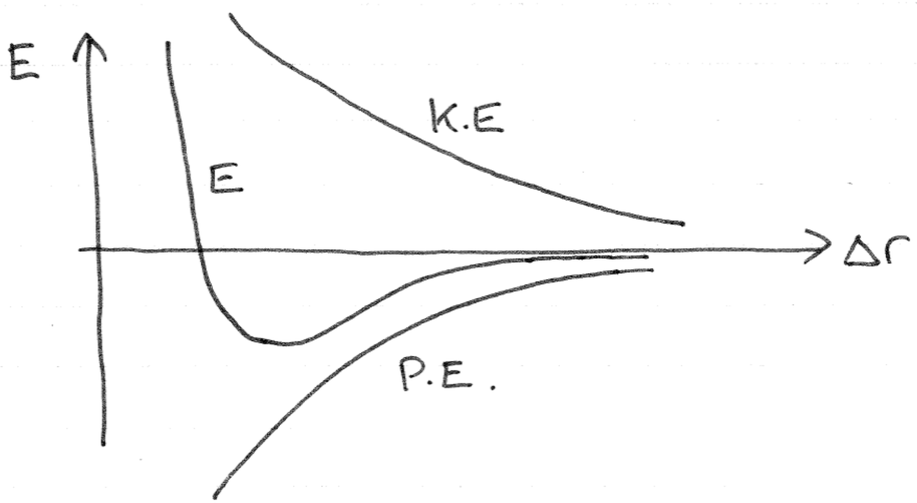
$$E \approx \frac{\hbar^2}{2m(\Delta r)^2} - \frac{Ze^2}{4\pi\varepsilon_0 \Delta r}$$



Minimize E :

$$\frac{dE}{d(\Delta r)} = 0 = \frac{-2\hbar^2}{2m(\Delta r)^3} + \frac{Ze^2}{4\pi\varepsilon_0(\Delta r)^2}$$

$$\Rightarrow \quad \Delta r \sim \frac{\hbar^2}{m}\left(\frac{4\pi\varepsilon_0}{Ze^2}\right) \equiv \text{ Bohr radius}$$



I recommend: https://www.feynmanlectures.caltech.edu/III_02.html#Ch2-S4

Features of Quantum Mechanics

1. probabilistic
2. no trajectories
3. gives rise to energy quantization

Properties of atoms §40.1

- There are FOUR (4) quantum numbers describing electrons in atoms
- Atoms have both *angular momentum* and *magnetism*
- Every electron whether *trapped* or *free* has a *spin angular momentum* and *spin magnetic dipole* – 'intrinsic' properties like electric charge

Principal quantum number	$n = 1, 2, 3...$	Distance from nucleus
Orbital quantum number	$\ell = 0$ to $(n-1)$	Orbital angular momentum
Orbital magnetic quantum number	$m_\ell = -\ell$ to ℓ	z-component of orbital angular momentum
Spin magnetic quantum number	$m_s = +1/2$ or $-1/2$	z-component of spin angular momentum

Multi-Electron Atoms §40.4-40.6

Pauli Exclusion Principle:

"No two electrons confined to the same trap can have the same set of values for their quantum numbers"

Each orbital has space for 2 electrons, with m_s values with opposite sign

$n = 1, 2, 3...$

$\ell = 0$ to $(n-1)$

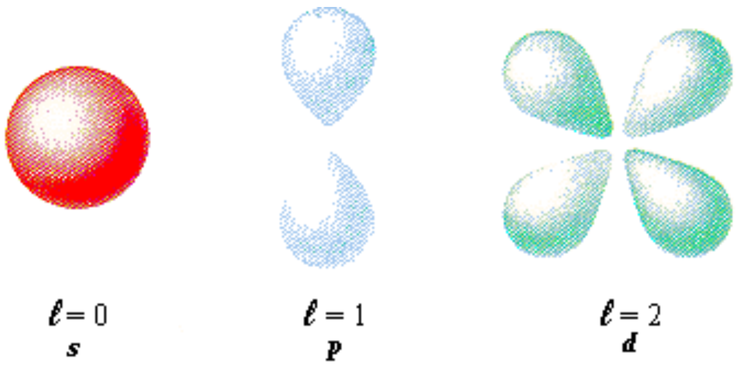
$m_\ell = -\ell$ to ℓ

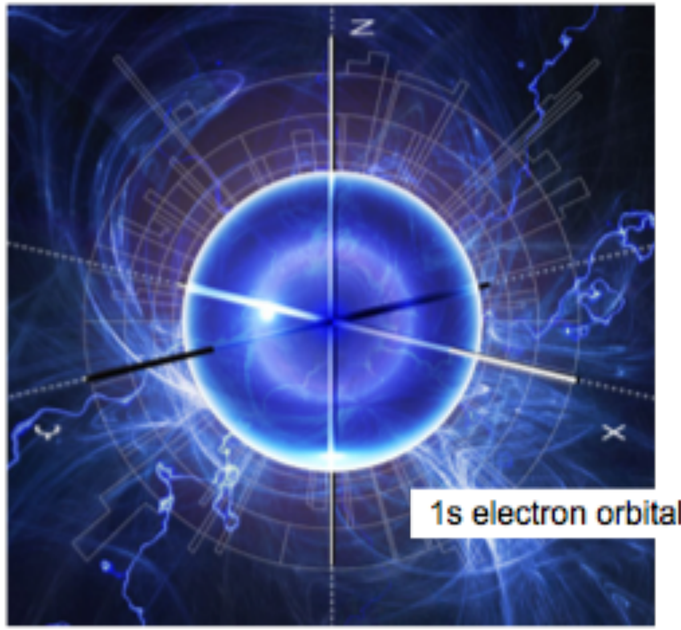
$m_s = +1/2$ or $-1/2$

Pauli Exclusion Principle

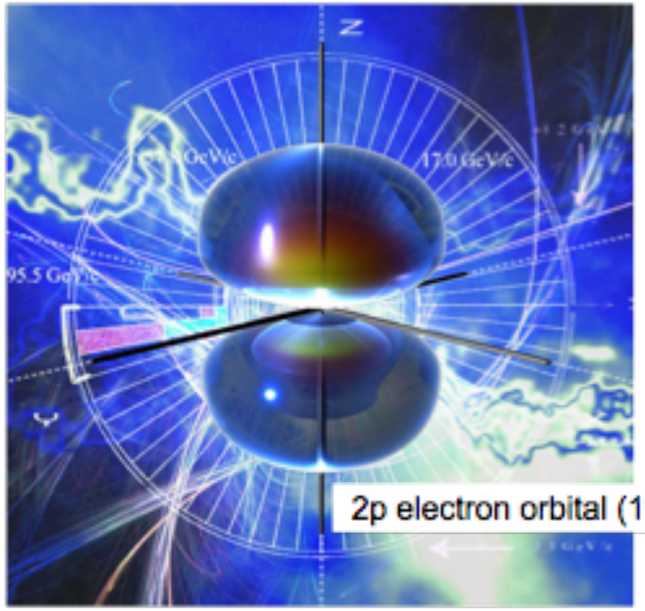
https://www.youtube.com/watch?v=Aoi4j8es4gQ

Electron Orbits



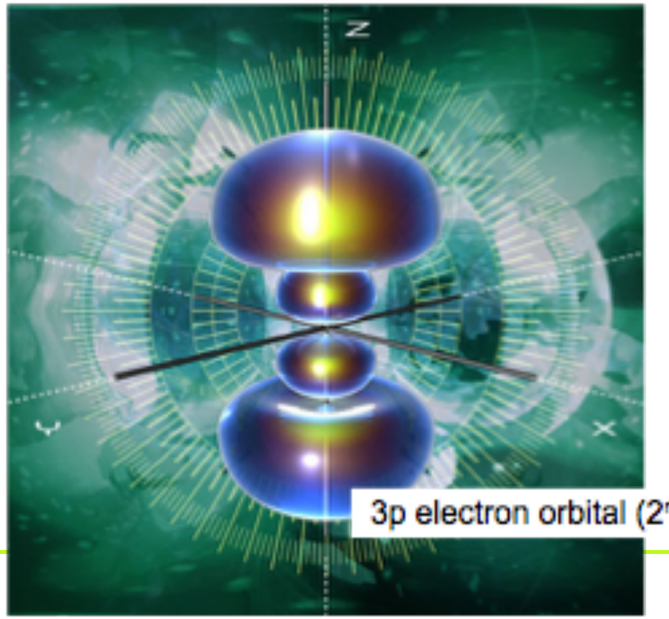


1s electron orbital (ground state)

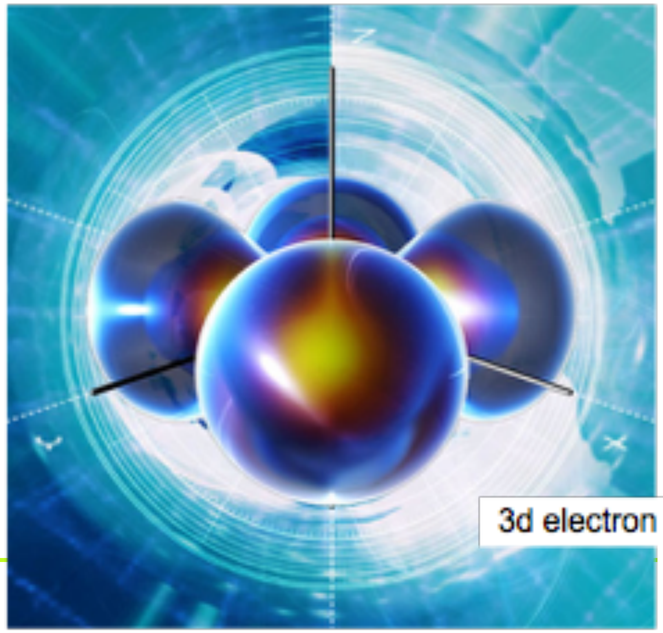


2p electron orbital (1st ex

'n' gives size of orbital
'l' gives shape of orbital



3p electron orbital (2nd excited state)



3d electron orbi

Quantum Mechanics & the Schrödinger Equation §38.7

Bohr theory – hybrid of Classical Mechanics and Quantum concepts
– worked for H atom, but not for many electron atoms

Proper theory: Quantum Mechanics (mid 1920's)
Schrödinger, Heisenberg, Dirac, Born, Jordan.

Matter wave: $\psi(x)$ "wave-function"

Schrödinger Equation (time-independent version) for a <u>free particle</u> in 1-D:

$$\boxed{\frac{d^2\psi}{dx^2} + k^2\psi = 0}$$

$$k = \frac{2\pi}{\lambda} = \frac{2\pi p}{h} = \frac{2\pi\sqrt{2m(E-U)}}{h}$$

angular wave number

Most general solution is: $\psi(x) = Ae^{ikx} + Be^{-ikx}$ <u>wave solutions</u>.

Schrödinger equation

- All the information about a system is contained in Schrödinger equation (1926)
- We *can't* know perfectly position and momentum (for example) but we do know other things (like energy levels and wavefunction) as it is in the SE

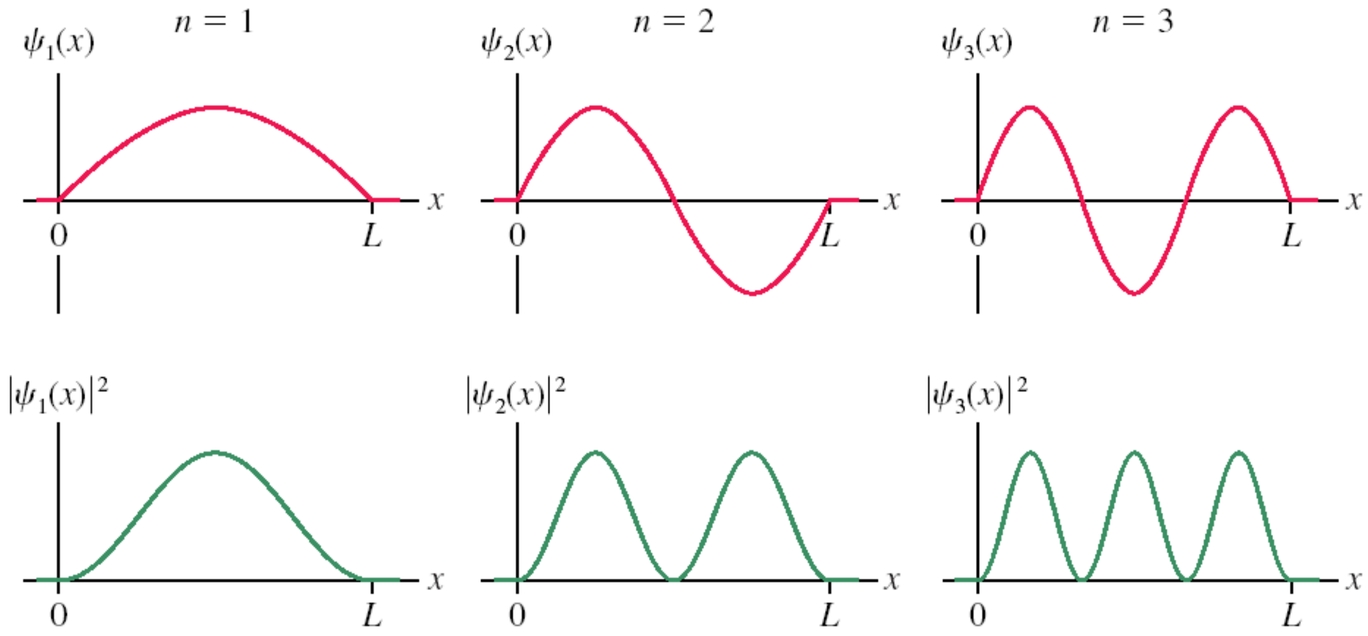
- Not so weird to have a wave equation:
  - Sound waves and waves on strings: Newtonian Mechanics
  - Light waves: Maxwells Equations
  - Matter waves: Schrödingers Equation

Probability interpretation (Born)

A particle does not have a specific location until its location is actually measured

$$|\psi|^2 = \text{probability density for position of particle}$$

FIGURE 41.7 Wave functions and probability densities for a particle in a rigid box of length L.



Success of Schrödinger's Eqn:

S.E. ➔ Heisenberg's Uncertainty Principle

$$\Delta x \Delta p_x \geq \hbar / 2 \qquad \Delta y \Delta p_y \geq \hbar / 2 \qquad \Delta z \Delta p_z \geq \hbar / 2$$

S.E. ➔ Bohr's formula for the energy levels of hydrogen with $V = -k \frac{q_1 q_2}{r^2}$

S.E. ➔ successful for many electron atoms and all known non-relativistic quantum systems.

Schrödingers Cat ‘thought experiment’

- 11 years after he first proposed his equation, after corresponding with Einstein, Schrödinger came up with this thought experiment
- A cat, a flask of poison & a sealed radioactive source are in a box…

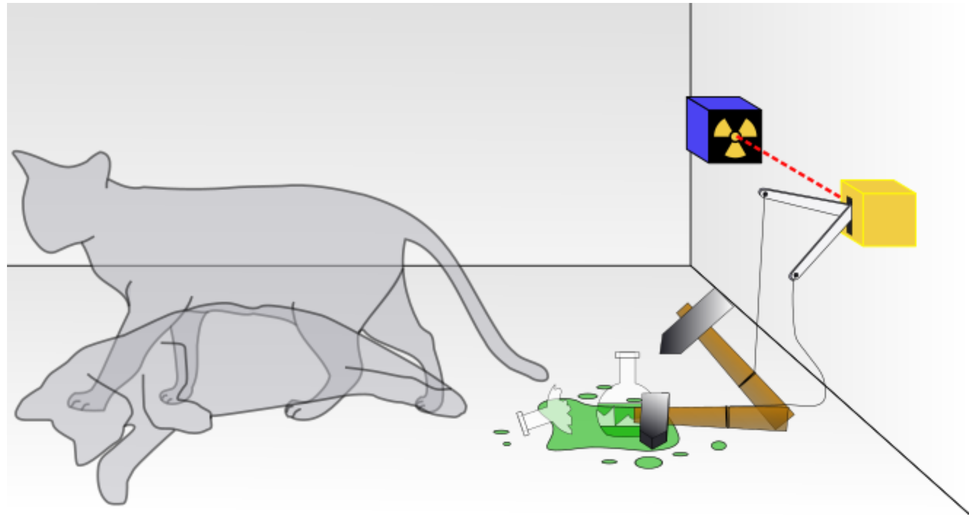
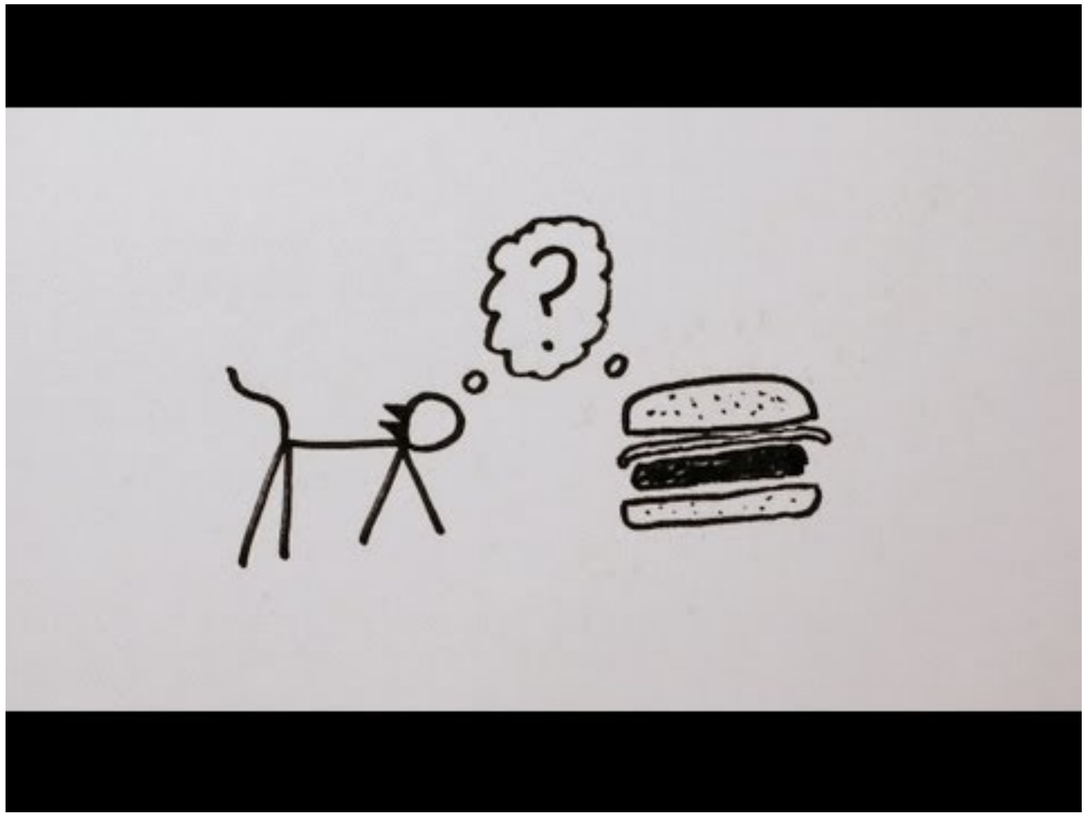


Image: wikimedia commons



Video: MinutePhysics

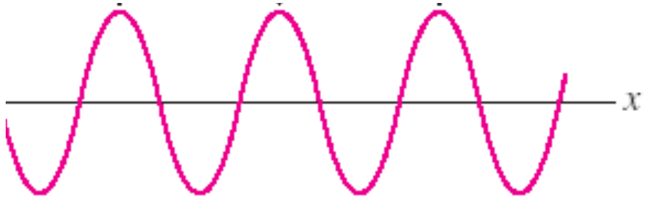
- There are multiple *interpretations* of quantum mechanics, some regard the “alive and dead” superposition as “real” and others do not

Wave Packets

Wave-particle duality described by a “wave-packet”

Consider a single wave:

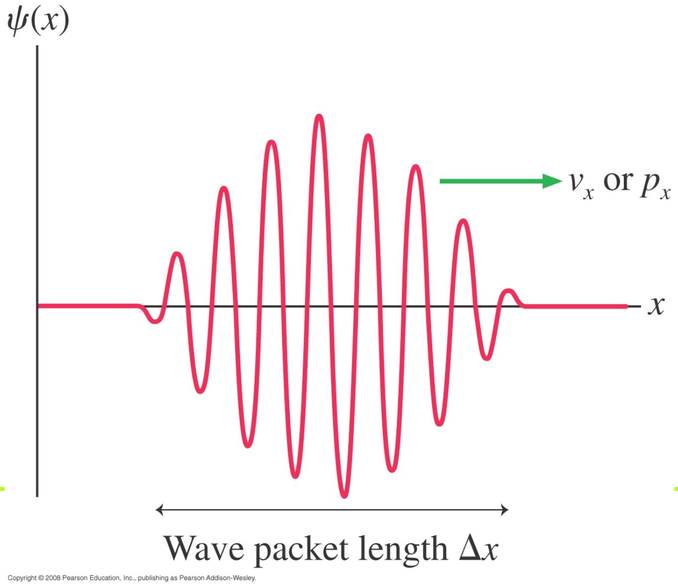
$$\Psi = e^{i(kx-\omega t)}$$



Momentum = $\hbar k$ exactly.
Position = anywhere b/w $\pm\infty$

A wave-packet is formed by adding many waves of different frequency (momentum).

$$\Psi = \int dk \phi(k) e^{i(kx-\omega t)}$$



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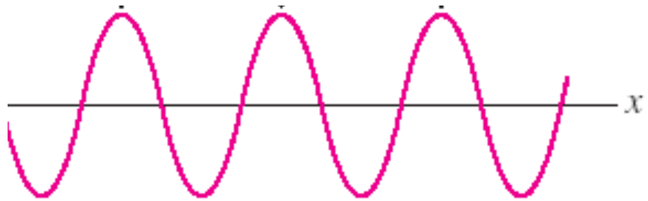
Uncertainty Relations revisited:

Uncertainty Principle is illustrated by simple wave properties.

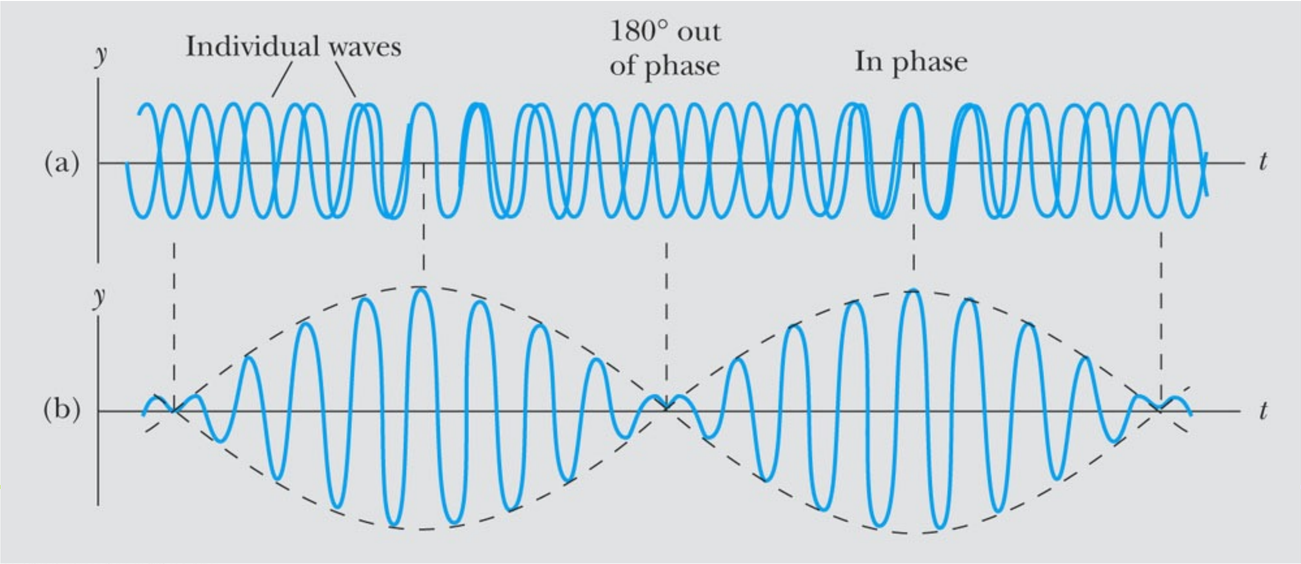
1-wave



$$\psi = A\cos[kx - \omega t]$$

$$\Delta x = \infty \quad \Delta p_x = 0$$

2-waves



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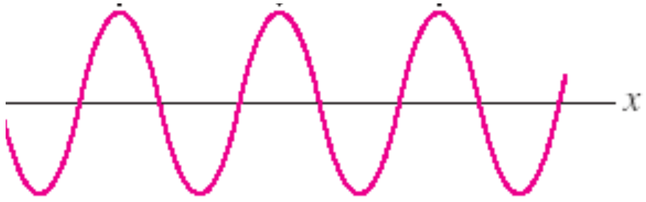
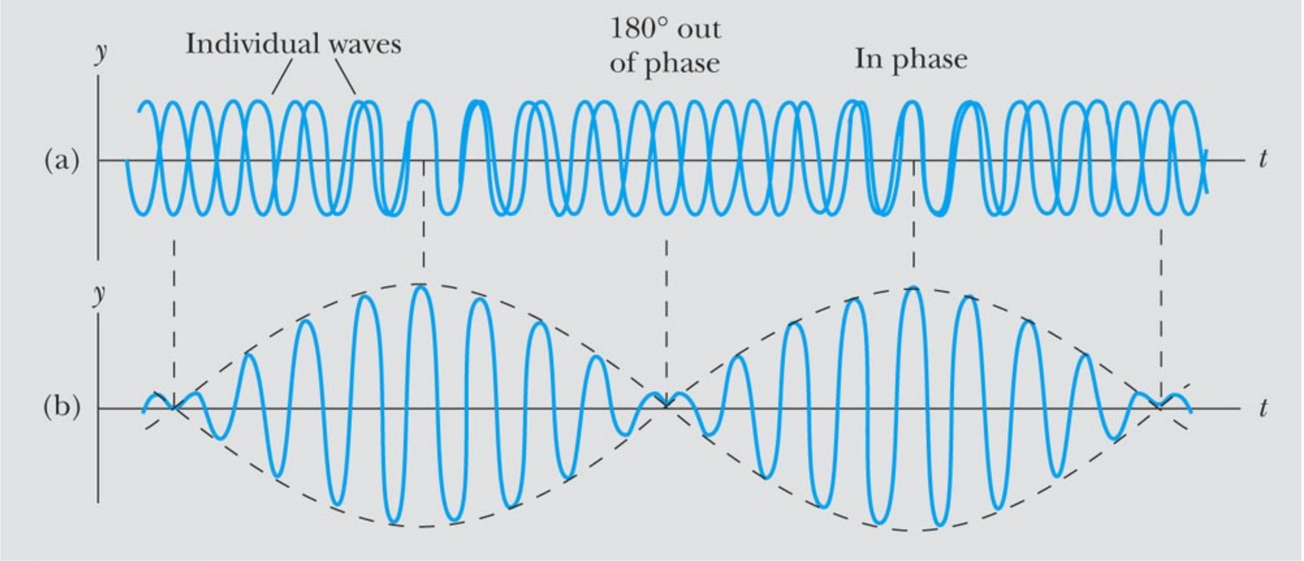
Beats

1-wave



2-waves



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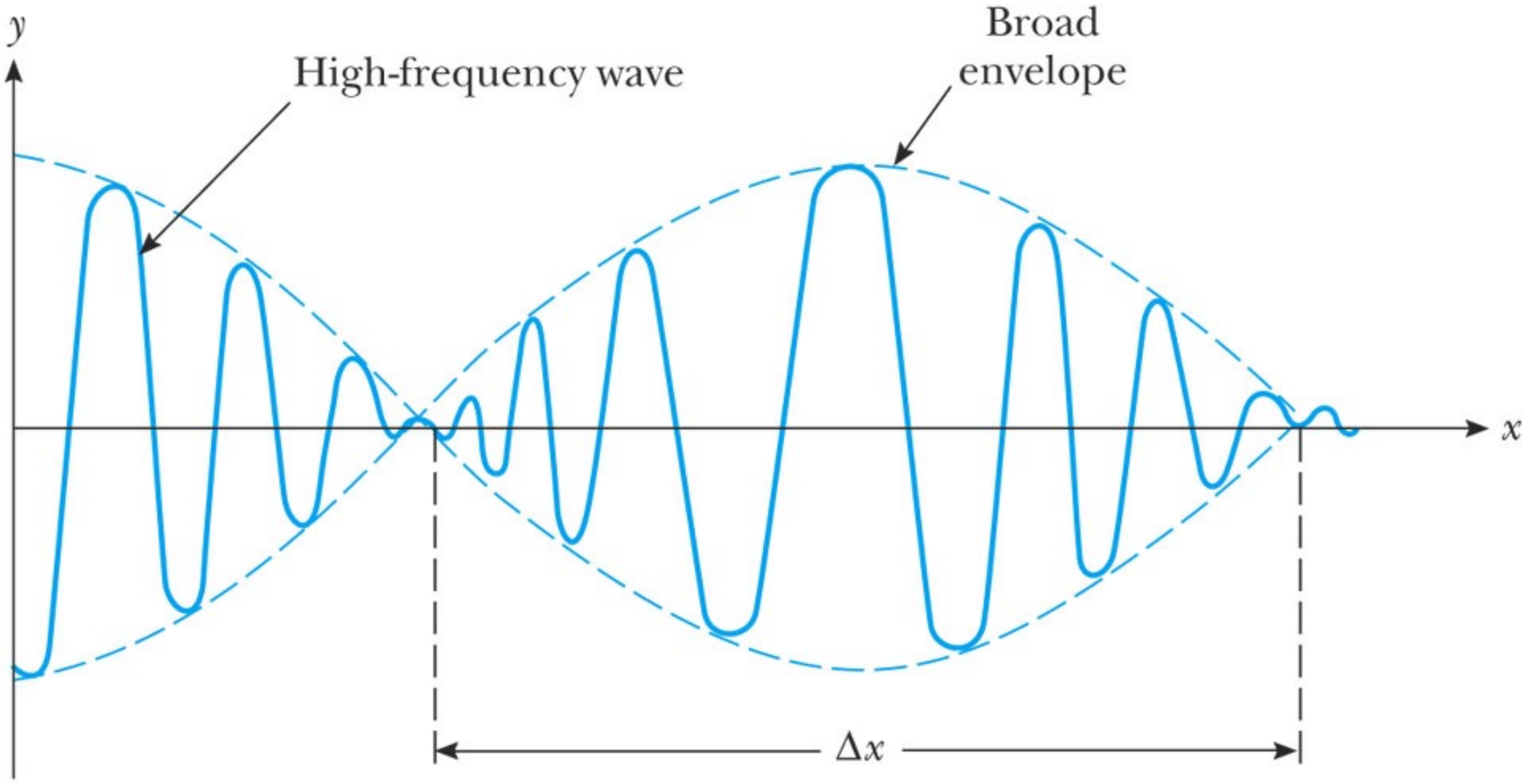
Beats

Adding two waves with slightly different frequencies:



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Schrödingers Equation

6 The probability that a particle is in a given small region of space is proportional to: **LO38.6**

(A) its energy.

(B) its momentum.

(C) the magnitude of its wave function.

(D) the wavelength of its wave function.

(E) the square of the magnitude of its wave function.

Summary:

The energy of a confined particle is quantized

$$E_n = n^2 \frac{h^2}{8mL^2} \qquad n = 1,2,3$$

Electrons in an atom have no trajectory

Wave Packets

Size of hydrogen atom

The correspondence principle